PAYTM BUILD FOR INDIA AI HACKATHON | MUMBAI EDITION

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FinSense - Financial Services Copilot

TRACK: AI-POWERED FINANCIAL JOURNEYS

UNIFIED AI INTELLIGENCE LAYER

CONSUMER CONTEXT

Payment habits, spend insights, personal financial management & guided actions.

→

PAYTM COPILOT

Intent understanding, real-time action synthesis & personalized orchestration.

→

MERCHANT ACTIONS

Automated cash flow, targeted customer offers & guided business tools.

BRIDGING CONTEXT TO ACTION

Transforming raw payment context into intelligent, proactive financial journeys for consumers & merchants.

100%

CONTEXTUAL PRECISION

DUAL-SIDE IMPACT

Consumers & Merchants unified in one ecosystem.

MUMBAI AI HACKATHON

FINTECH INNOVATION

FRICTION POINTS

Digital financial journeys suffer from significant friction despite available payment infrastructure. Persistent context gaps prevent seamless execution across consumers and merchants.

CONSUMER FRICTION

01. Product Discovery

High search friction and difficulty finding relevant financial products tailored to immediate needs.

02. Policy & Loan Terms

Opaque, complex terms create drop-off during understanding and evaluation phases.

03. Navigation Friction

Disjointed user interfaces lead to multi-step drop-offs across payment journeys.

MERCHANT FRICTION

Fragmented Ecosystem

Merchants face severe operational fragmentation across multiple key touchpoints:

- **Lending:** Disconnected credit options.
- **Insurance & Claims:** Manual workflows.
- **Settlements:** Delayed reconciliation.
- **Payment Issues:** High failure resolution costs.

THE CONTEXT GAP

A massive disconnect exists between user intent and financial offerings.

Current State: Users receive generic, untargeted promotions.

Missing Need: Actionable, context-aware financial assistance at the exact moment of transaction.

Core Impact

Despite robust underlying payment infrastructure, the lack of real-time contextual intelligence creates widespread drop-offs for consumers and heavy operational friction for merchants across the entire financial lifecycle.

SHARED AI INTELLIGENCE

Streamlined pipeline powering dual experiences

CORE ARCHITECTURE

One unified intelligence layer driving context-aware journeys across customer touchpoints.

5-STEP PIPELINE

Process Flow

01	02	03	04	05
Detect Signal	**Understand Context**	**Recommend Journey**	**Guide Execution**	**Complete**

DUAL IMPACT

Personalized real-time execution

CONSUMER EXPERIENCE

Trigger: Authorized transactions like vehicle service payments

Automated Sequence:

01. Vehicle insurance discovery

02. Verified policy comparison

03. Seamless application handoff

2X

SHARED LAYER

Contextual intelligence adapting to user role in real time

MERCHANT EXPERIENCE

Trigger: Analyzing payment and settlement trends and other signals

Automated Sequence:

01. Tailored lending options

02. Insurance guidance & claims assistance

03. Rapid issue resolution

02

SYSTEM ARCHITECTURE & TRUST

Hybrid AI orchestration with deterministic Java/SQL validation and granular user privacy controls.

FRONTEND LAYER

React & Tailwind UI

Client-side interface driving guided completion, dynamic user flows, and real-time state feedback.

CORE ORCHESTRATION

Java Spring Boot / Python FastAPI

Handles workflow state, system business logic, deterministic calculations, and secure API routing.

STORAGE

PostgreSQL

Relational data persistence and validation logging.

LOCAL INFERENCE

Ollama LLM

On-device AI inference for natural context interpretation.

INTEGRATION

Simulated Partner APIs

Extensible endpoint connectors for external data verification and partner integrations.

Banks/NBFC and other partners can list themselves and have tiers.

TRANSPARENCY GUARANTEE

EXPLAINABLE RECOMMENDATIONS

Every output provides auditable rationale and step-by-step guided completion, eliminating "black box" decisions and boosting user trust.

GOVERNANCE FRAMEWORK

PRIVACY & USER CONTROLS

- **Explicit Consent Model**
  Requires clear user approval before processing sensitive data inputs.
- **Strict Data Minimization**
  Processes only payload attributes necessary for specific calculations.
- **Frequency Caps & Customization**
  Strict limits on recommendation triggers with customizable user preferences.

CORE ARCHITECTURAL USP

HYBRID AI + DETERMINISTIC BUSINESS RULES

Blends AI flexibility for context interpretation with strict Java/SQL code paths for accurate calculations and state validation. Ensures 100% mathematical correctness.

STAKEHOLDER VALUE & IMPACT

01 / CONSUMER BENEFITS

Streamlined User Experience

- **Reduced Discovery Effort:** Faster access to relevant financial products and services.
- **Clearer Policy Information:** Transparent terms and coverage details provided upfront.
- **Guided Navigation:** Intuitive step-by-step assistance throughout every user journey.

02 / MERCHANT ADVANTAGE

Unified Operations Hub

- **Single Interface:** Centralized dashboard for managing all partner operations seamlessly.
- **Multi-Product Management:** Integrated handling of lending, insurance, and claims.
- **Issue Resolution:** Direct workflows for handling payment and operational queries.

03 / PAYTM ECOSYSTEM IMPACT

Platform Optimization

- **Discovery Relevance:** Smarter algorithms connecting users to targeted offerings.
- **Higher Journey Completion:** Reduced drop-offs across conversion funnels.
- **Support Efficiency:** Decreased customer support overhead and ticket volumes.

01. RELEVANCE

Recommendation Relevance

Evaluates accuracy and engagement of tailored suggestions.

02. COMPLETION

Completion Rates

Tracks successful end-to-end journey conversions.

03. EFFICIENCY

Time-to-Completion Steps

Measures reduction in user effort and total duration.

04. RETENTION

Dismissal & Opt-Outs

Monitors friction points and user rejection frequency.

Business Model & Hackathon MVP

Commercial & Business Model

01 Distribution & Referral Revenue

Monetization is driven through strategic partner distributions and referral revenue streams, scaling effectively alongside partner growth.

02 Operational Savings

Emphasizes core operational efficiency to significantly reduce overhead costs while increasing automated processing capabilities.

03 Suitability-First Approach

Prioritizes user fit over commercial margins.

Ensures long-term customer trust and retention by matching users to ideal financial products rather than maximizing short-term commission payouts.

Hackathon MVP Scope

A Shared Recommendation Engine

Features an integrated engine processing synthetic consumer payment records to drive seamless, automated insurance handoffs.

B 6-Month Synthetic Data Analysis

Evaluates six months of synthetic merchant payment records to extract actionable behavioral analytics and risk profiles.

C Trend-Driven Guidance

Integrated Lending, Insurance & Claims

Delivers intelligent, automated recommendations across lending options, targeted insurance policies, and proactive claims support based on real-time trends.

CORE PLATFORM CAPABILITIES

01

Contextual Discovery Engine

Categorizes real-time payment context to proactively suggest relevant financial products.

Eliminates search friction with high-precision moment-of-need triggers.



02

Explainable Advisory Layer

Local Ollama LLM generates transparent natural-language rationales for recommendations.

Translates complex terms into plain language for confident user decision-making.

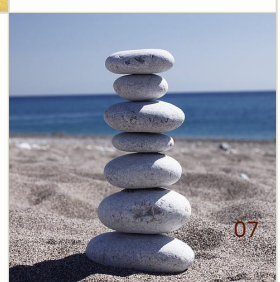


03

Deterministic Guardrails & Hub

Java Spring Boot and SQL enforce calculations, state rules, consent, and frequency caps.

Unifies merchant lending, claims guidance, and issue resolution in one workflow.



THANK YOU



FinSense: Bridging financial context to intelligent, actionable financial journeys for consumers and merchants.

TEAM SLYTHERINS

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Paytm Build for India AI Hackathon | Mumbai Edition